Heat Stress Days in Corn Production

Definition

1. **Heat stress days (HSDs):** Periods where daily maximum temperatures exceed the threshold at which corn growth and yield are negatively affected.
2. In corn, heat stress is critical during **June–August** (reproductive stage).
3. **Key thresholds:**
    a. Yields decline once temperatures exceed **86 °F (30 °C)** (Nielsen, 2020).
    b. Four or more consecutive days in the mid-90s °F can reduce yields by **~1% per day**; losses accelerate after the fifth or sixth day (Hoegemeyer, 2011).

Known Effects on Corn

- **Photosynthesis:** Net photosynthesis in corn is optimized at ~86 °F; rates decline sharply above this temperature (Hopkins, 1999).
- **Reproduction:** Heat stress during pollination reduces kernel set and grain fill.
- **Water stress:** Higher temperatures increase transpiration demand and VPD, forcing stomatal closure, which reduces CO_2 uptake and photosynthesis.
- **Yield impacts:** Extended heat stress during reproductive stages is strongly correlated with reduced harvestable yield (Crafts-Brandner & Salvucci, 2002; Waqas et al., 2021).

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Measurement Systems for Heat Stress

1. Stress Degree Days (SDD)

- **Definition:** The difference between the daily maximum temperature and the optimum maximum temperature (86 °F for corn).
- **Example:** If daily maximum = 92 °F → 6 SDD accumulated.
- **Thresholds:** When cumulative SDD > 140, yields typically fall **below trend** (Taylor, 2012).
- **Application:** Simple, corn-specific metric to track heat stress intensity and duration.

2. Growing Degree Days (GDD)

- **Definition:** Measures cumulative heat for crop development.
- **Formula:** $GDD=(Tmax+Tmin)2-TbaseGDD=2(Tmax+Tmin)-Tbase$, where base = 50 °F for corn.
- **Use:** Predicts growth stages and harvest readiness.

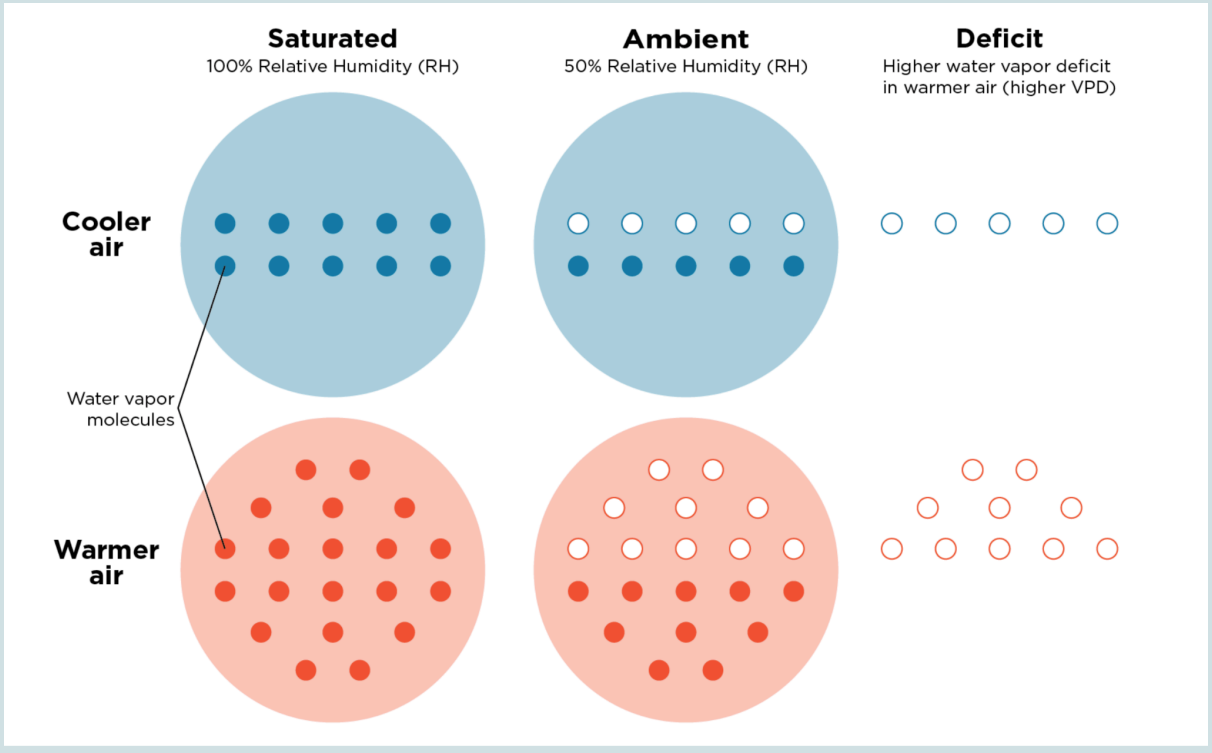
- **Limitations:**
    - Does not account for extreme heat thresholds (plateaus max temp at 86 °F).
    - Ignore water stress and precipitation impacts.
    - Useful for phenology, **not yield under heat extremes**.

3. Wet-Bulb Globe Temperature (WBGT)

- Composite index of temperature, humidity, radiation, and wind.
- Common in occupational health; less directly applied to crop yield but relevant for plant-atmosphere interactions.

4. Vapor Pressure Deficit (VPD)

- **Definition:** Difference between actual atmospheric moisture and saturation capacity at a given temperature.
- **Impact:** High VPD accelerates water loss from plants, intensifying heat and drought stress.
- **Mechanism:** Higher air temperatures increase VPD exponentially, even if relative humidity remains constant (Hsiao, 1999).



Both ambient circles show 50% relative humidity (holding half as much water as they would at saturation), however they show different VPDs as warmer air/temperatures need more moisturization to reach saturation while cooler temperatures require less.

“When plants perform photosynthesis, they open their pores to exchange oxygen for carbon dioxide. But, because the inside of a leaf typically has a much higher concentration of water than the surrounding air, opening their pores also means losing water via **diffusion** to the atmosphere.”

“Since photosynthesis is a do or die process for most plants (shout out to the parasitic plants of the world), changes to the outside environment can have serious consequences. When VPD is high (meaning the air *could* hold a lot more moisture), water loss from plants occurs more quickly, since the difference in concentration between the leaf and air is particularly pronounced. Over multiple weeks and months, a consistently *thirsty atmosphere*, pulling water out of plants, can lead to dead or, at a minimum, really dry vegetation.”

Plants perform photosynthesis by opening their pores to exchange oxygen for carbon dioxide. Typically, plant’s leaves hold much higher water concentration than the surrounding air. Therefore, performing photosynthesis can mean a loss of water for the plant via diffusion to the atmosphere.

When VPD is high (meaning the air could hold a lot of moisture…which is when temperatures are hot), water loss from plants occur much quickly

consequences of high VPD = over multiple weeks and months, a plant in a constantly thirsty environment will be dead, or at minimum, really dry vegetation.

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5. Other Experimental Indices

- **Heat Index (HI):** Incorporates temperature and humidity, but rarely applied directly to crops.
- **Extreme Heat Degree Days (EHDD):** Proposed refinements to track cumulative effects of very high temperatures.
- **Crop-Specific Indices:** New work on **Heat Stress Index (HSI)** and **Temperature Condition Index (TCI)**integrates remote sensing with thermal metrics (see ScienceDirect, 2023).

Table 2. Estimated corn yield loss when drought stress persists for four or more consecutive days. (Drought stress indicated when the uppermost, fully expanded leaf was visibly wilted.)

Corn Growth Stage	Estimated Yield Loss per Day of Stress (%)
Early vegetative (VE - V12)	1 – 3
Late vegetative (V12 to VT)	2 – 5
Pollination to Blister (R2)	3 – 9
Milk (R3)	3 – 6
Dough (R4)	3 – 5
Dent (R5)	2 – 4
Maturity (R6)	0

Licht, M. and S. Archontoulis. 2017. Influence of Drought on Corn and Soybean. Iowa State Univ. Extension.

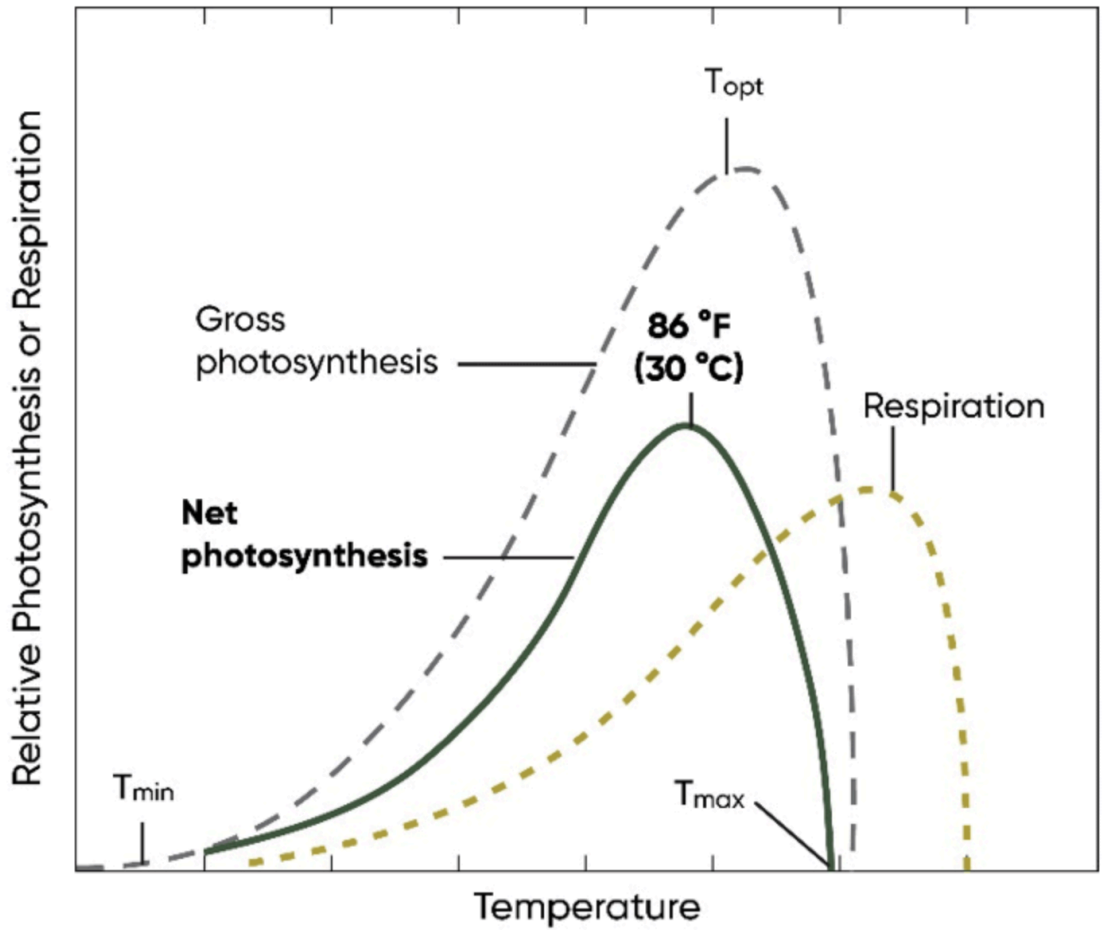


Figure: Generalized model of temperature effects on rates of gross photosynthesis, respiration, and net photosynthesis. Net photosynthesis in corn is optimized at 86 °F. (Figure adapted from Hopkins, 1999).

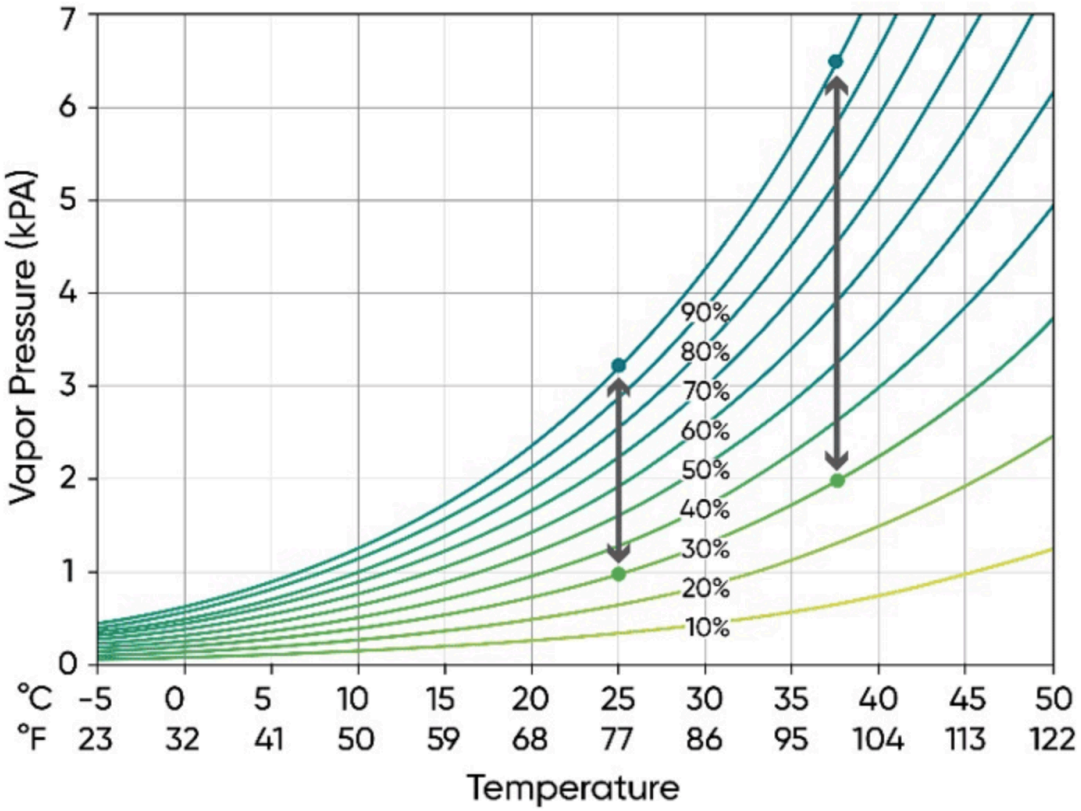


Figure 7: Vapor pressure for water by relative humidity and temperature.

“The greater the vapor pressure deficit between the leaf interior and the surrounding air, the faster the rate at which water will be pulled out of the plant and evaporated. Temperature is important to this equation because VPD increases exponentially with increasing temperature, even if RH stays constant.”

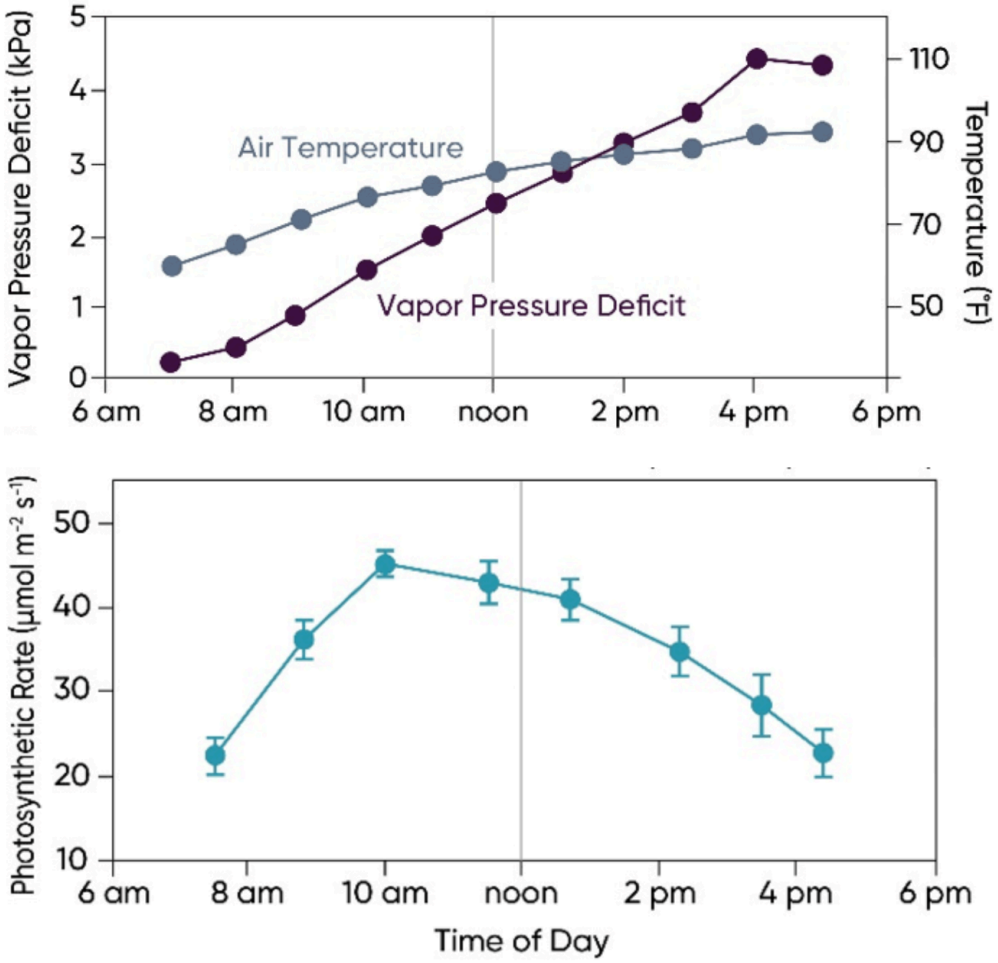


Figure : Air temperature, atmospheric vapor pressure deficit, and leaf photosynthetic rate in irrigated corn over the course of a day (Hirasawa and Hsiao, 1999).

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Temperature Condition Index (TCI)

- Remote sensing–based indicator designed to measure plant stress caused by excessive temperature.
- TCI is derived from satellite land surface temperature anomalies, typically scaled 0–100, where low values indicate high temperature stress.
- Can be integrated with NDVI for dual assessment of drought and heat stress on crops.
- Application: Useful for regional-to-global monitoring of crop stress, complements GDD/SDD which rely on ground-level weather data.

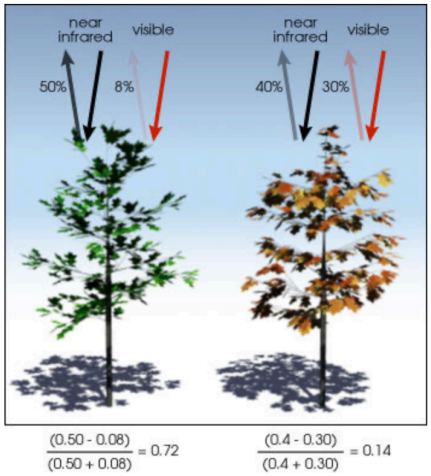


Image courtesy of NASA.

$$NDVI = \frac{(NIR - Red)}{(NIR + Red)}$$

higher NDVI means healthier plant

Heat Impact Score (HIS)

- Recently developed composite index to assess the cumulative impact of extreme heat events on crop growth.
- Incorporates frequency, intensity, and duration of extreme heat days relative to crop-specific thresholds.
- Application: Can be tailored to corn by weighting critical growth stages (e.g., tasseling/pollination) more heavily in the scoring system.
- Advantage: Moves beyond simple daily thresholds to a multi-dimensional risk metric.

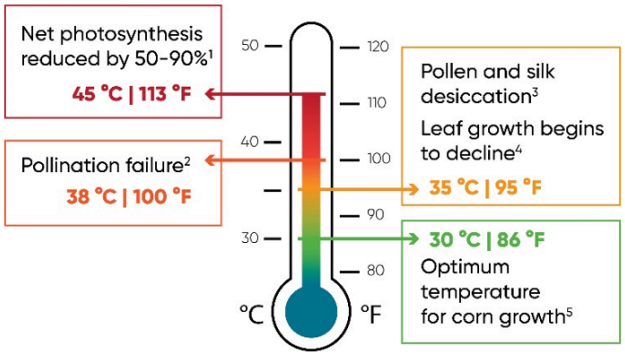
Minimum and maximum cardinal temperatures in crop and pest management and use of information.

Crop or pest	minimum	maximum	information use
Corn	**50°F**	**86°F**	**crop development**
Soybean	50°F	86-90°F	crop development
Black cutworm	50°F	--	300 DD from egg to cutting
Stalk borer	41°F	--	predicting migration
Bean leaf beetle	46°F	--	2nd generation emergence
Seedcorn maggot	39°F	--	seed treatment on replant
Alfalfa weevil	48°F	--	larval presence in fields
Western bean cutworm	50°F	--	adult emergence/ egglaying

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Application to Corn Growing

- **Yield Forecasting:**
  - SDD provides a direct, crop-specific measure of cumulative heat stress.
  - When SDD > 140, risk of sub-trend yields is high (Taylor, 2012).
- **Management Decisions:**
  - Identifying stress windows helps schedule irrigation, fertilization, and hybrid selection.
  - Remote sensing (NDVI) combined with GDD/SDD can improve yield prediction (Becker-Reshef et al., 2010).
- **Climate Change Context:**
  - Increasing frequency of HSDs in June–August may lead to systemic yield declines without adaptation.
  - Integration of SDD, VPD, and NDVI in predictive models is critical for early-warning systems.



- Higher temperatures cause the transpiration rate of plants to increase, placing a greater demand on soil water supply and potentially accelerating the onset of drought stress.
- Corn plants respond to water stress by closing their stomata, which helps preserve water but also reduces the rate at which plants are able to take in the CO_2 needed for photosynthesis.

The tool displays estimates of actual cumulative GDD from planting to today's date, then estimates of cumulative GDD for the remainder of the season.
link to device
link to NDAWn
link to crop moisture index

look more into this web: https://www.sciencedirect.com/science/article/abs/pii/S0168192323003179
Research about TCI and HIS index (heat)